Normalization and its use in AI

Definition:

Normalization is a specific form of feature scaling when a dataset has features of varying ranges. It is the process of adjusting data to a common scale without distorting the relationships between values, so that features with different ranges can be treated equally by machine learning models.

Importance of normalization:

Normalized data enhances model performance and improves the accuracy of a model. It aids algorithms that rely on distance metrics, such as k-nearest neighbors or support vector machines, by preventing features with larger scales from dominating the learning process.

Normalization fosters stability in the optimization process, promoting faster convergence during gradient-based training. It mitigates issues related to vanishing or exploding gradients, allowing models to reach optimal solutions more efficiently.

Normalized data is also easy to interpret and thus, easier to understand. When all the features of a dataset are on the same scale, it also becomes easier to identify and visualize the relationships between different features and make meaningful comparisons.

Normalization Techniques:

Min-Max scaling and Z-score normalization (standardization) are the two fundamental techniques for normalization.

1- Min-Max Scaling:

Min-max scaling is very often simply called normalization or linear normalization. It transforms features to a specified range, typically between 0 and 1.

The formula for min-max scaling is:

$$Xscaled = \frac{X - Xmin}{Xmax - Xmin}$$

Where:

X is the original value
X min is the minimum value of the feature
X max is the maximum value of the feature

Advantages:

- It ensures that all features have the same scale.
- It is simple to implement and the results are easy to interpret.

Disadvantages:

- It is sensitive to outliers, the extreme values can distort scaling which makes most data points cluster near 0 or 1.
- Fixed range limits flexibility for datasets with changing scales.

Example 1:

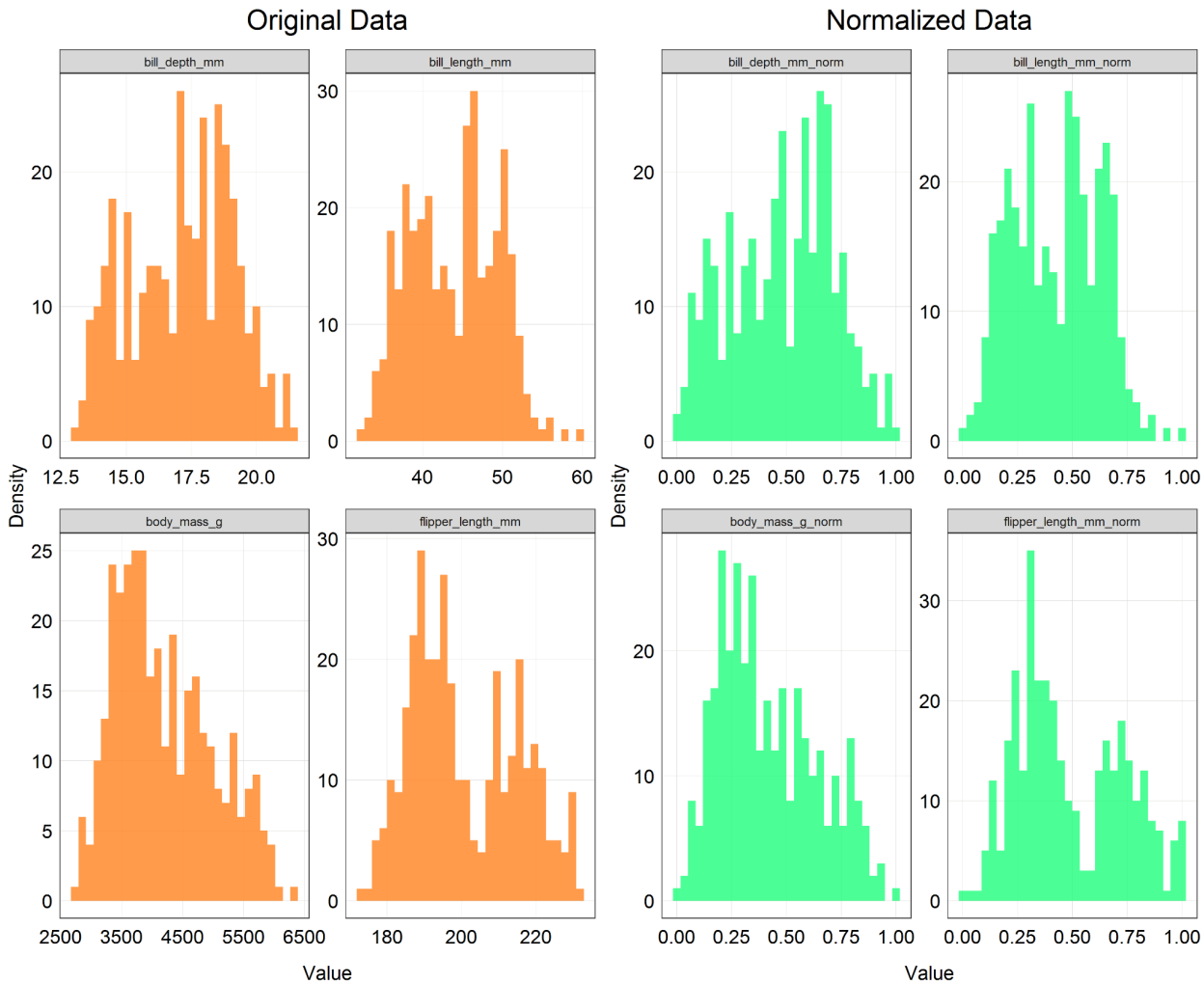

Figure (1)

As illustrated in figure (1) original data that varied on a large scale is normalized to a scale of [0 , 1] remaining the same relations between original values.

2- Z-score normalization (standardization):

Z-score normalization (standardization) assumes a Gaussian (bell curve) distribution of the data and transforms features to have a mean (μ) of 0 and a standard deviation (σ) of 1.

The formula for min-max scaling is:

$$z = \frac{x - \mu}{\sigma}$$

Where:

z = standardized value
x = original feature value
μ = mean of the feature
σ = standard deviation of the feature

This technique is particularly useful when dealing with algorithms that assume normally distributed data, such as many linear models. Unlike the min-max scaling technique, feature values are not restricted to a specific range in the standardization technique. This normalization technique basically represents features in terms of the number of standard deviations that lie away from the mean.

Advantages:

- Useful when features follow a Gaussian (normal) distribution.
- Works well with algorithms assuming data is centered (Linear/Logistic Regression, PCA, SVM).
- Less affected by outliers compared to MinMaxScaler.

Disadvantages:

- Does not preserve the original distribution of data (values become relative to mean/std).
- If the data is not close to normal distribution, results may not be optimal.
- Still sensitive to extreme outliers (though less than MinMaxScaler).

Example 2:

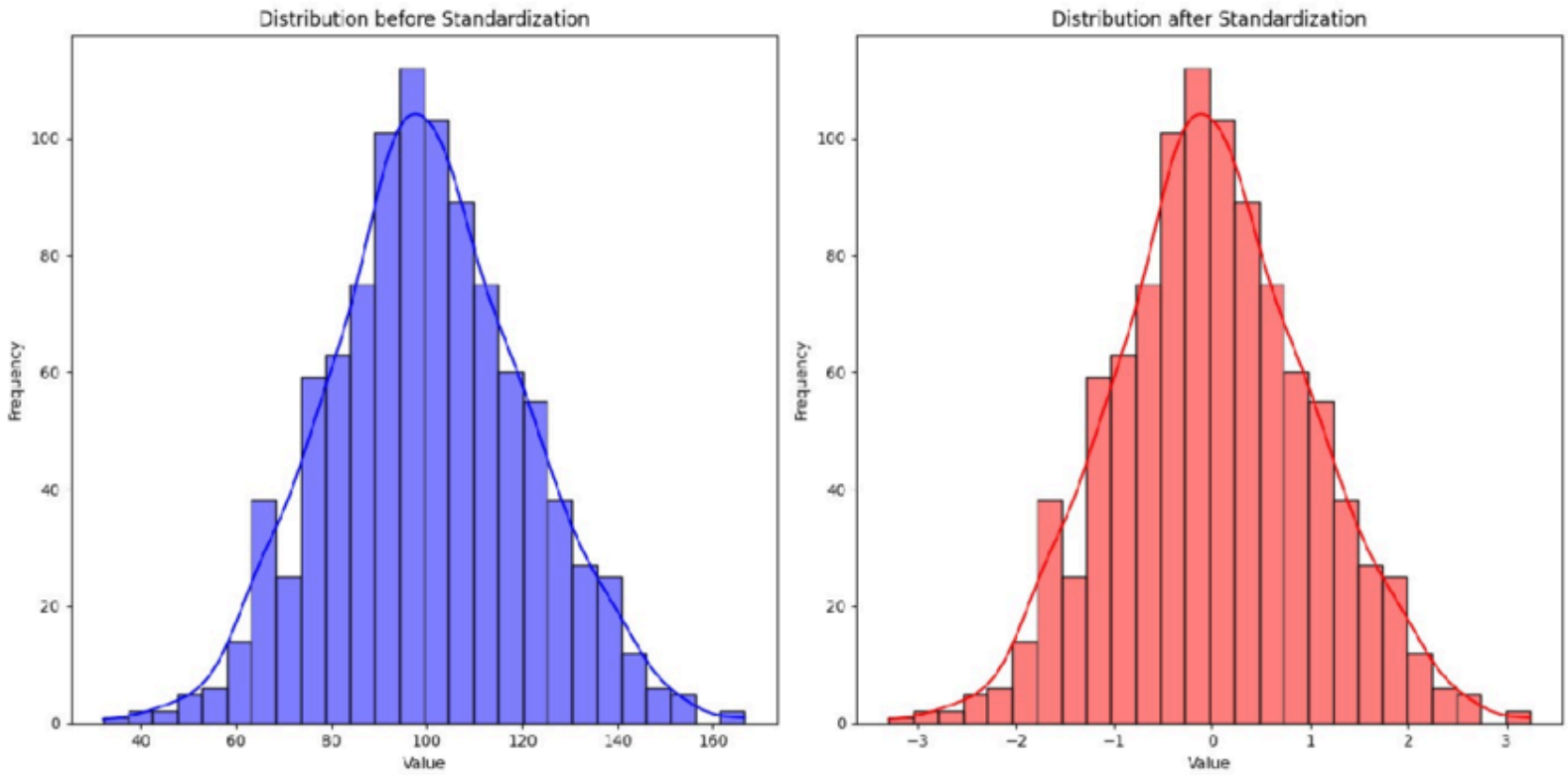

Figure (2)

As illustrated in figure (2) unlike the linear normalization, the standardized data is not bounded between a specific range between 0 and 1. This technique basically represents features in terms of the number of standard deviations that lie away from the mean.

Matrix Normalization Examples:

Original Matrix:

$$\begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

1) Min–Max Normalization (Range [0,1])

- **Step 1:** Find Minimum and Maximum values.

Minimum = 2

Maximum = 8

- **Step 2:** Apply the formula:

X_norm = (X - X_min) / (X_max - X_min)

X_norm = (X - 2) / 6

- **Step 3:** Apply to each element:

(2 - 2)/6 = 0

(4 - 2)/6 = 0.33

(6 - 2)/6 = 0.67

(8 - 2)/6 = 1

Final Normalized Matrix:

$$\begin{bmatrix} 0 & 0.33 \\ 0.67 & 1 \end{bmatrix}$$

2) Z-Score Normalization (Standardization)

- **Step 1:** Compute the Mean (μ).

$\mu = (2 + 4 + 6 + 8) / 4 = 5$

- **Step 2:** Compute Standard Deviation (σ).

$(2 - 5)^2 = 9$

$(4 - 5)^2 = 1$

$(6 - 5)^2 = 1$

$(8 - 5)^2 = 9$

Variance = 20 / 4 = 5

$\sigma = \sqrt{5} \approx 2.24$

- **Step 3:** Apply the formula:

$X_norm = (X - \mu) / \sigma$

$(2 - 5)/2.24 \approx -1.34$

$(4 - 5)/2.24 \approx -0.45$

$(6 - 5)/2.24 \approx 0.45$

$(8 - 5)/2.24 \approx 1.34$

Final Standardized Matrix:

$$\begin{bmatrix} -1.34 & -0.45 \\ 0.45 & 1.34 \end{bmatrix}$$